

Trainings ▾

Failure Analysis Inspection

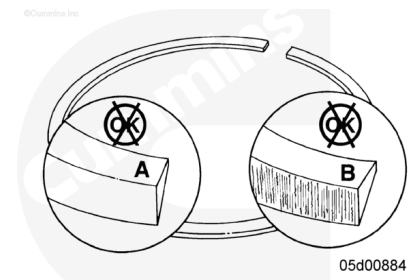
Inspect the piston rings for the following:

- Abrasive wear.

Note : Abrasive wear of the intermediate ring can be indicated by a rapid reduction of the dark finish coating on the front face of the ring, in some cases, to the point where the dark finish coating is no longer visible (A). This is commonly referred to as full face ring wear. This rapid reduction will typically leave a sharp edge on the bottom of the intermediate ring. Abrasive wear can also be indicated by concentrated vertical scratches on the top ring (B).

Abrasive wear can be caused by:

1. Ingested abrasive material
2. Inadequate cleaning during a previous repair
3. Particles embedded in the bore
4. High soot content in the lubricating oil from extended oil drain intervals
5. Scuffing and scoring.

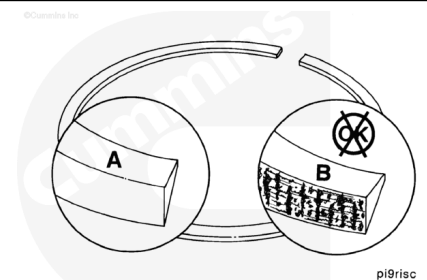


Note : Scuffing and scoring is indicated by heavy scratches, metal discoloration, and voids (B).

Scuffing and scoring can be caused by:

1. Engine overheating
2. Oil dilution
3. Improper maintenance of the lubrication system
4. Piston cooling nozzle malfunction
5. Oil ring plugged by deposits.

Note : Scuffing and scoring on the piston rings indicates a breakdown of the oil film on the cylinder bore wall, causing transfer of material from the piston ring face to the cylinder bore.



Note : Oil ring plugging is indicated by deposits on the oil ring grooves (B).

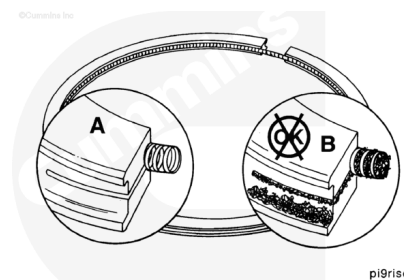
Oil ring plugging can be caused by:

1. Low engine operating temperatures - long periods of idling or a cooling system malfunction
2. Extended oil change intervals
3. Use of the wrong grade of engine oil
4. Use of a poor quality engine oil.

Note : Plugging of the oil ring drains restricts oil drain back, which floods the piston ring belt area, resulting in a loss of oil control.

Note : The following measurements are intended for inspecting new piston rings.

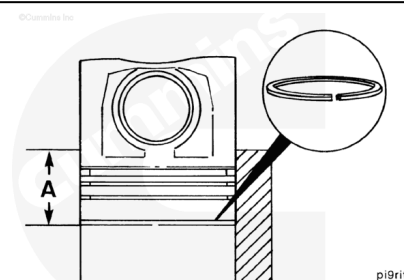
Note : Before completing this inspection, make sure the cylinder bore is within specification. Refer to Procedure 001-026 (</qs3/pubsys2/xml/en/procedures/40/40-001-026.html>) in Section 1.



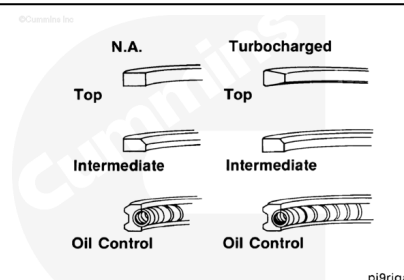
Measure

Front Gear Train

Measure the piston ring gap by installing the piston rings into the cylinder bore in which they will be used. Position the rings below the ring reversal area. Position each ring in the cylinder 89 mm [3.5 in] below the top deck (A), and using a clean piston to square it with the bore.



The piston ring sets are different. While both sets consist of three rings, the top ring of the turbocharged/aftercooled set has a keystone profile that operates in a ni-resist insert cast into the piston. The naturally aspirated (NA) top ring is square cut and operates in a groove machined into the aluminum piston.

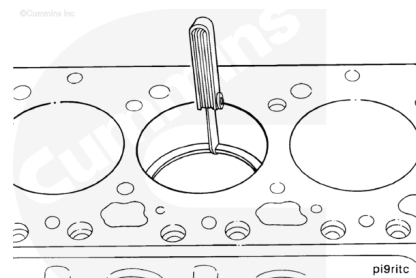


Note : The top ring for a turbocharged engine is not the same as the top ring for a naturally aspirated engine.

Use a feeler gauge to measure the gap.

B3.9, B4.5, and B5.9 Ring Gap

	mm		in
Oil Control	0.25	MIN	0.010
	0.55	MAX	0.022



Piston Ring Gap Cross Referencing Tables

- To establish the correct top and intermediate piston ring gap, the part numbers of the piston rings must be known. If the engine has been rebuilt previously, then check for records of the parts used during the previous rebuild. If no records are present for the parts used during rebuild, then this table can't be used to determine the correct ring gap. If this is the case, then new piston rings should be used during the rebuild of the engine.
- The cross referencing tables are ordered by ring size (Standard, 0.5 mm oversize, then 1.0 mm oversize) and then by part number in descending order. First identify the correct bore / piston ring size, then find the part number of the rings in the appropriate table.

Note : Piston ring gap sizes are given in mm [inch].

Standard Size Piston Ring

Top ring			Intermediate Ring		
Part Number	Minimum Gap	Maximum Gap	Part Number	Minimum Gap	Maximum Gap
4897899	0.22 [0.009]	0.38 [0.015]	4897900	0.60 [0.024]	0.86 [0.034]
3959079	0.26 [0.010]	0.42 [0.017]	3897719	0.25 [0.010]	0.61 [0.024]
3947678	0.26 [0.010]	0.42 [0.017]	3943544	0.85 [0.033]	1.21 [0.048]
3937392	0.35 [0.014]	0.51 [0.020]	3942265	0.25 [0.010]	0.61 [0.024]
3918315	0.40 [0.016]	0.76 [0.030]	3940777	0.25 [0.010]	0.61 [0.024]
3904529	0.40 [0.016]	0.76 [0.030]	3937393	0.55 [0.022]	0.90 [0.035]
3902401	0.40 [0.016]	0.76 [0.030]	3932519	0.25 [0.010]	0.61 [0.024]
2831144	0.30 [0.012]	0.46 [0.018]	3904531	0.25 [0.010]	0.61 [0.024]
3897673	0.40 [0.016]	0.70 [0.028]	3904530	0.25 [0.010]	0.61 [0.024]

Standard Size Piston Ring					
Top ring			Intermediate Ring		
Part Number	Minimum Gap	Maximum Gap	Part Number	Minimum Gap	Maximum Gap
			3902400	0.25 [0.010]	0.61 [0.024]
			3902286	0.25 [0.010]	0.61 [0.024]
			3897719	0.25 [0.010]	0.55 [0.022]

0.5 mm [0.02 in] Oversize Piston Ring					
Top ring			Intermediate Ring		
Part Number	Minimum Gap	Maximum Gap	Part Number	Minimum Gap	Maximum Gap
4898855	0.22 [0.009]	0.38 [0.015]	4898858	0.60 [0.024]	0.86 [0.034]
3957244	0.26 [0.010]	0.40 [0.016]	3943545	0.85 [0.033]	1.21 [0.048]
3937428	0.35 [0.014]	0.51 [0.020]	3942266	0.25 [0.010]	0.61 [0.024]
3919050	0.40 [0.016]	0.76 [0.030]	3937430	0.55 [0.022]	0.90 [0.035]
3902392	0.40 [0.016]	0.76 [0.030]	3933583	0.25 [0.010]	0.61 [0.024]
2831145	0.30 [0.012]	0.46 [0.018]	3904347	0.25 [0.010]	0.61 [0.024]
3866105	0.40 [0.016]	0.70 [0.028]	3902393	0.25 [0.010]	0.61 [0.024]
			3866107	0.25 [0.010]	0.55 [0.022]

1.0 mm [0.04 in] Oversize Piston Ring					
Top ring			Intermediate Ring		
Part Number	Minimum Gap	Maximum Gap	Part Number	Minimum Gap	Maximum Gap
4898856	0.22 [0.009]	0.38 [0.015]	4898859	0.60 [0.024]	0.86 [0.034]
3957245	0.26 [0.010]	0.40 [0.016]	3943546	0.85 [0.033]	1.21 [0.048]

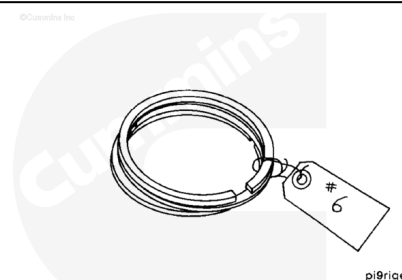
1.0 mm [0.04 in] Oversize Piston Ring

Top ring			Intermediate Ring		
Part Number	Minimum Gap	Maximum Gap	Part Number	Minimum Gap	Maximum Gap
3937429	0.35 [0.014]	0.51 [0.020]	3942267	0.25 [0.010]	0.61 [0.024]
3919051	0.40 [0.016]	0.76 [0.030]	3937431	0.55 [0.022]	0.91 [0.036]
3902441	0.40 [0.016]	0.76 [0.030]	3933585	0.25 [0.010]	0.61 [0.024]
2831146	0.30 [0.012]	0.46 [0.018]	3904348	0.25 [0.010]	0.61 [0.024]
3866106	0.40 [0.016]	0.70 [0.028]	3902442	0.25 [0.010]	0.61 [0.024]
			3866108	0.25 [0.010]	0.55 [0.022]

If the piston ring gap is **not** within specification:

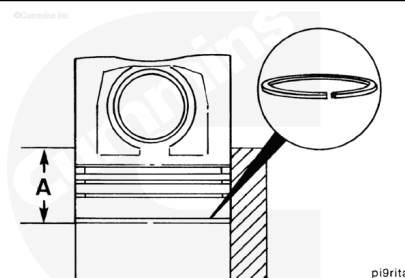
1. Verify the correct type and part number piston ring is being used
2. Verify the cylinder bore is within specification. Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-026.html)
3. Verify the piston ring gap measurement is being taken 89 mm [3.5 in] below the cylinder block deck
4. Try another set of piston rings.

After each ring gap is checked and verified, identify the ring sets for installation in the cylinder bore where the end gap was measured.



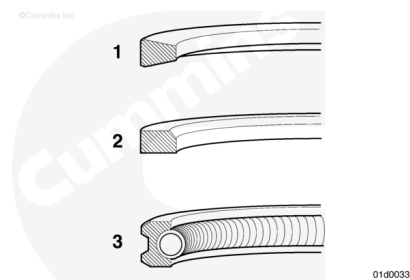
Rear Gear Train

Measure the piston ring gap by installing the piston rings into the cylinder bore in which they will be used. Position the rings below the ring reversal area. Position each ring in the cylinder 89 mm [3.5 in] below the top deck (A), and using a piston to square it with the bore.



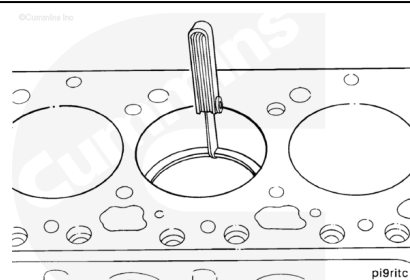
The piston ring type and location can be identified by piston ring profile.

1. Top piston ring
2. Intermediate piston ring
3. Oil control ring.



Use a feeler gauge to measure the gap.

B4.5 RGT Engines Ring Gap			
	mm		in
Top	0.30	MIN	0.012
	0.46	MAX	0.018
Intermediate	0.82	MIN	0.032
	1.18	MAX	0.047
Oil	0.22	MIN	0.010
	0.58	MAX	0.023



If the piston ring gap is **not** within specification:

1. Verify the correct type and part number piston ring is being used
2. Verify the cylinder bore is within specification. Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-026.html)
3. Verify the piston ring gap measurement is being taken 89 mm [3.5 in] below the cylinder block deck
4. Try another set of piston rings.

After each ring gap is checked and verified, identify the ring sets for installation in the cylinder bore where the end gap was measured.

